

Introduction to PyTorch

Tensors, Computation Graphs, Autograd, Ecosystem, and Framework Choice

Abstract

PyTorch is an open-source, Python-oriented library for tensor computation and deep learning. Its central design connects multidimensional data, differentiable operations, neural-network components, optimization, hardware acceleration, distributed execution, and export tools without hiding ordinary Python control flow. These notes build a beginner-first mental model of that system, trace its practical evolution, correct several time-sensitive deployment details, compare PyTorch with TensorFlow through project constraints, and provide a verified numerical autograd example.

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1 Purpose and prerequisite vocabulary

Modern generative artificial intelligence applications—including large language models—are usually built from neural networks trained on large collections of examples. Several terms must be separated before PyTorch can be placed correctly:

Artificial intelligence (AI) is the broad goal of constructing systems that perform tasks associated with intelligent behavior.

Machine learning (ML) is an approach in which a system learns patterns from data rather than relying only on hand-written rules.

Deep learning is a branch of machine learning that uses neural networks with multiple transformation stages, usually called *layers*.

Generative AI refers to models that generate new content such as text, images, audio, or code.

Large language model (LLM) is a neural network trained to model sequences of language tokens at large scale.

Library is reusable software that exposes data structures and operations. PyTorch is a library; it is not itself a trained model.

A useful learning order is

tensors $\longrightarrow$ tensor operations $\longrightarrow$ computation graphs
 $\longrightarrow$ automatic differentiation $\longrightarrow$ models, losses, and optimizers
 $\longrightarrow$ deployment and scaling.

Each arrow denotes a prerequisite relationship. Tensors hold data, operations transform it, computation graphs record dependencies, automatic differentiation computes gradients, and optimizers use those gradients to change model parameters.

PyTorch: from data to a usable model

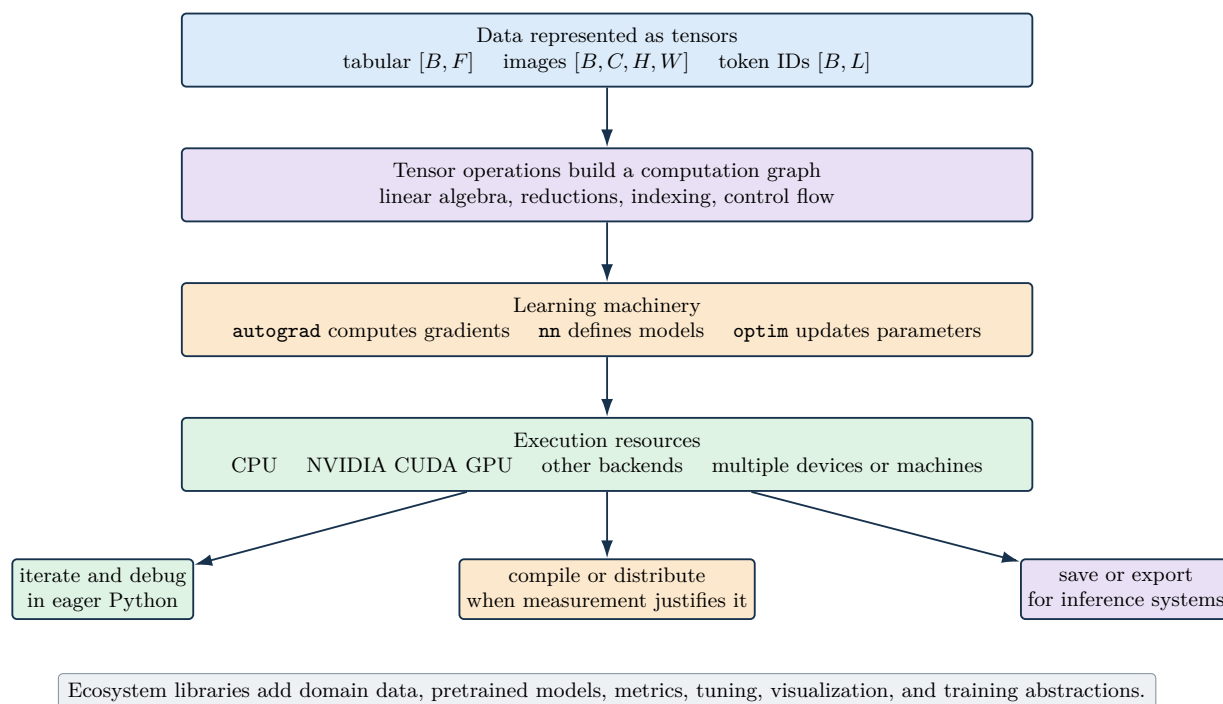


Figure 1: PyTorch connects tensor data, differentiable computation, learning components, hardware, and model delivery. [Official reference](#).

Intuition

PyTorch is best understood as a stack of responsibilities rather than as one opaque “AI engine.” A tensor can be useful without a neural network; a model can run without training; compilation can improve execution without changing the mathematical model; and exporting a program is different from saving its learned parameter values.

2 What PyTorch is

Definition

PyTorch is an open-source, Python-oriented library that provides multidimensional tensors, mathematical operations, automatic differentiation, neural-network building blocks, optimization algorithms, hardware backends, distributed-training tools, and program-export facilities.

The name joins two ideas. *Torch* supplied a scientific-computing tradition centered on tensor operations and GPU execution, while *Py* emphasizes the Python interface and Python ecosystem. The practical combination mattered because many data workflows already used Python libraries such as NumPy and SciPy. A researcher or engineer could therefore add differentiable tensor computation without first adopting Lua, the language used by the older Torch framework.

The word *open-source* means that the implementation can be inspected, modified, and redistributed under its license. It does not mean that every model, dataset, accelerator, or cloud service used with PyTorch is free or open.

2.1 Why deep learning needs tensor computation

A neural network repeatedly applies numerical transformations to arrays of values. A dense layer, for example, maps an input vector x to an output vector z :

$$z = Wx + b.$$

Here:

- $x \in \mathbb{R}^F$ is an input with F features;
- $W \in \mathbb{R}^{O \times F}$ is a matrix of learnable weights;
- $b \in \mathbb{R}^O$ is a learnable bias vector;
- $z \in \mathbb{R}^O$ is the layer output;
- $\mathbb{R}$ denotes real-valued numbers, and the superscripts state shapes.

For $F = 2$, $O = 1$, $x = [2, 3]^\top$, $W = [4, -1]$, and $b = 0.5$,

$$\begin{aligned} z &= 4(2) + (-1)(3) + 0.5 \\ &= 8 - 3 + 0.5 \\ &= 5.5. \end{aligned}$$

The calculation produces one scalar because $O = 1$. PyTorch supplies the tensor operations needed for this calculation and records enough information to differentiate it when training is required.

3 Tensors, axes, shapes, dtypes, and devices

Definition

A *tensor* is a multidimensional array with a shape, a numeric data type, and a device. An *axis* is one direction in that array, and a *shape* lists the length of every axis in order.

Common representations include:

Data item	Typical shape	Meaning
Tabular sample	$[F]$	F measured features for one row.
Mini-batch of rows	$[B, F]$	B rows, each with F features.
Color image	$[C, H, W]$	C channels, height H , and width W .
Image mini-batch	$[B, C, H, W]$	B images with the same image shape.
Token sequence	$[L]$	L integer token identifiers.
Sequence	$[B, L]$	B equally padded token sequences.
mini-batch		
Video clip	$[T, C, H, W]$	T frames, each with channels, height, and width.
Time series	$[T, F]$	T time steps with F values at each step.

The symbols are dimensions, not values: B is batch size, F is feature count, C is channel count, H is image height, W is image width, L is sequence length, and T is a time or frame count.

A tensor's *dtype* states how each value is represented. Examples include `float32`, `float16`, `bfloat16`, `int64`, and `bool`. A tensor's *device* states where its storage and operations live, such as CPU memory or an NVIDIA CUDA device.

Different data, one tensor abstraction

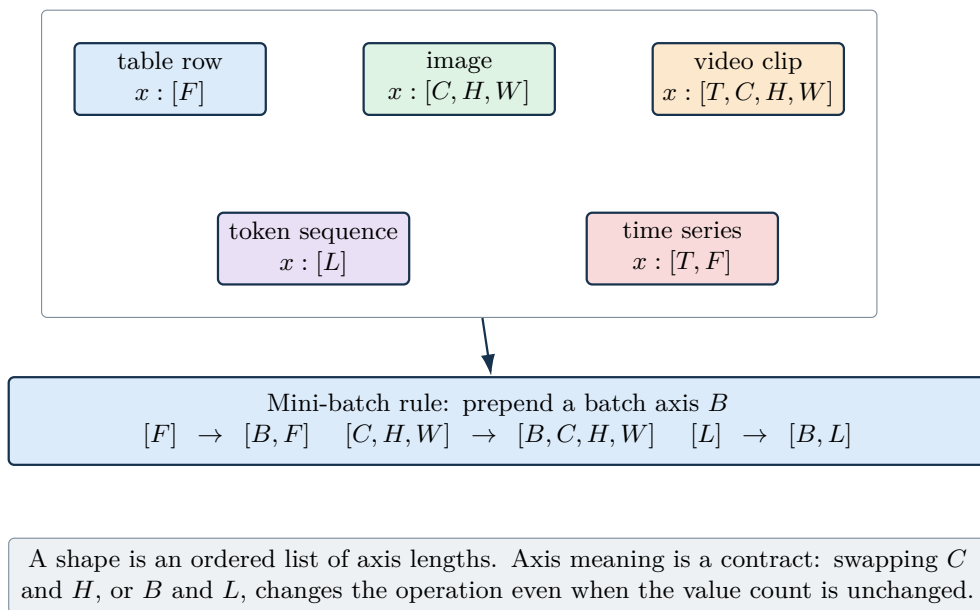


Figure 2: Tensor shapes encode the meaning and order of data axes; batching normally prepends a batch axis. [Official reference](#).

Shape check

Two tensors can contain the same number of values and still be incompatible. For example, $[B, C, H, W]$ and $[B, H, W, C]$ have equal value counts but different axis contracts. Similarly, a CPU tensor and a CUDA tensor cannot participate in an ordinary operation until they are placed on a compatible device.

4 How PyTorch evolved

4.1 From Torch to a Python interface

The older Torch framework demonstrated that tensor operations could be accelerated on GPUs and used to implement influential neural-network architectures. Its Lua interface nevertheless imposed adoption friction on a community whose data-science work increasingly centered on Python. PyTorch joined Torch-like tensor capability with a Pythonic interface and interoperability with existing Python tools.

4.2 PyTorch 0.1 and dynamic computation graphs

The early PyTorch release combined two especially important properties:

1. ordinary Python could express the model and surrounding control flow;
2. computation graphs were built dynamically as operations executed.

This combination made experimentation and debugging attractive in research settings. A user could inspect intermediate tensors with ordinary Python tools and could alter a model's executed path through normal conditionals or loops.

4.3 PyTorch 1.0 and the research–production bridge

PyTorch 1.0 placed greater emphasis on moving trained work beyond interactive research. TorchScript supplied a historical route for serializing and optimizing model logic, and Caffe2 integration brought production-oriented capabilities into the broader project. Subsequent 1.x releases expanded distributed training, ONNX interoperability, quantization, domain packages, community integrations, and cloud support.

4.4 PyTorch 2.x and optional compilation

PyTorch 2.0 introduced `torch.compile` as its main compilation-facing API while preserving eager-mode development. Compilation attempts to transform executable regions into more efficient code. It is optional: the same model can first be developed eagerly and then measured with compilation.

Two performance terms should not be confused:

Latency is the time required for one request or one batch.

Throughput is the amount of work completed per unit time, such as samples per second.

Lower latency and higher throughput are desirable, but a change can improve one without improving the other. Performance depends on model structure, tensor shapes, precision, hardware, data transfer, compilation overhead, and batch size.

A practical PyTorch timeline

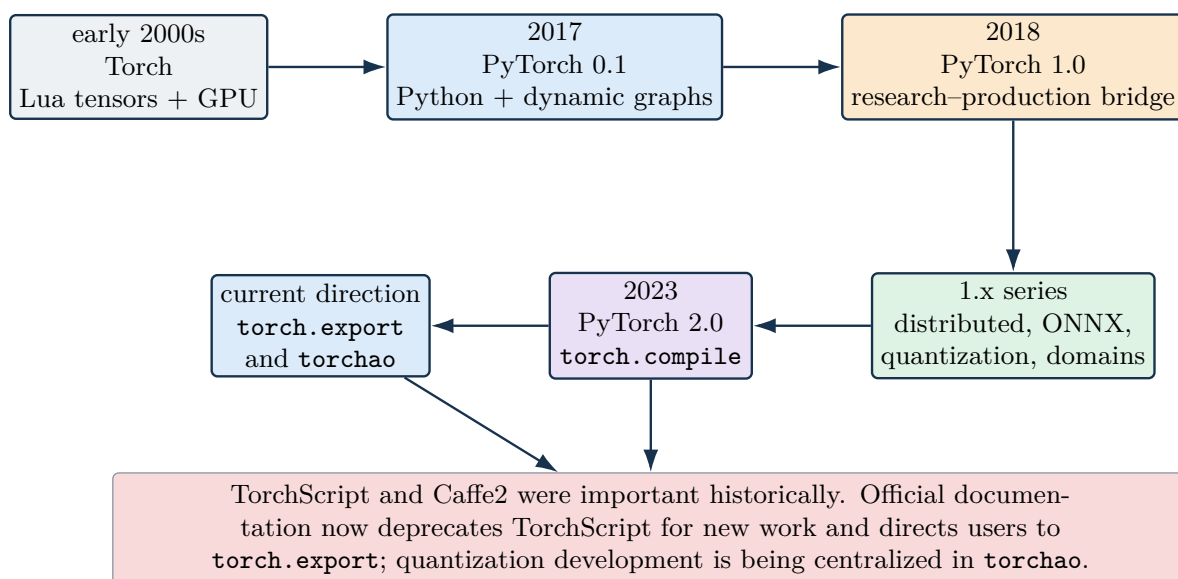


Figure 3: The PyTorch timeline moves from Torch and dynamic Python execution toward distributed tools, interoperability, and optional compilation. [Official reference](#).

Current-practice note

TorchScript remains historically important, but current official documentation marks it deprecated and recommends `torch.export` for new export work. Official quantization documentation also states that quantization development is being centralized in `torchao`. Historical module names should therefore be learned as context, not copied blindly into a new production design.

5 Computation graphs

Definition

A *computation graph* represents values and operations as a directed graph. A directed edge indicates that one value is needed to compute another. The graph records dependency, not merely visual layout.

Consider

$$y = (a + b)c.$$

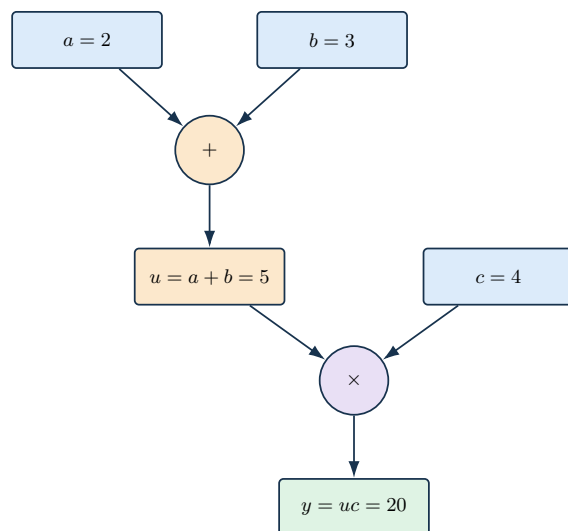
Introduce an intermediate value $u = a + b$. With $a = 2$, $b = 3$, and $c = 4$,

$$u = a + b = 2 + 3 = 5,$$

$$y = uc = 5 \cdot 4 = 20.$$

The expanded form exposes two operations and their order. The compact expression and expanded graph compute the same result.

Expanded computation: $y = (a + b)c$



Nodes store values or operations; arrows record dependencies. The graph states the calculation order: add first, then multiply.

$$\frac{\partial y}{\partial a} = 4, \quad \frac{\partial y}{\partial b} = 4, \quad \frac{\partial y}{\partial c} = 5.$$

Figure 4: The expression $y = (a + b)c$ becomes an addition node followed by a multiplication node, with dependencies preserved. [Official reference](#).

A neural network is also a composition of mathematical operations. A layer may compute $z = Wx + b$, apply an activation function $h(z)$, and feed the result to later layers. Recording this chain permits reverse-mode differentiation during backpropagation.

5.1 Static-style and dynamic graphs

A static-style workflow prepares a graph before ordinary execution. Changing the graph structure may require rebuilding, retracing, or recompiling it. A dynamic eager workflow executes Python first and

records the operations that actually occur. If a condition selects a different branch on a later input, the recorded graph can differ.

The road-map analogy is useful if interpreted carefully. A fixed approved route resembles a prepared graph: construction follows the map. If an unexpected obstacle appears, a dynamic process can choose a route around it while work proceeds. The analogy describes structural flexibility; it does not imply that dynamic execution is always faster.

Static plan versus dynamic execution

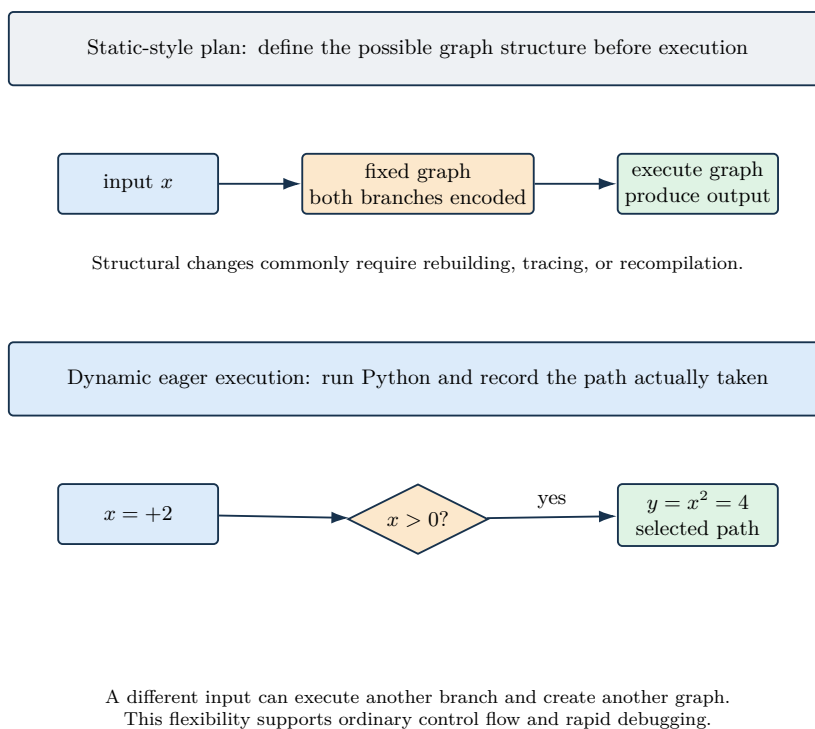


Figure 5: Static-style execution prepares possible structure, whereas eager dynamic execution records the control-flow path taken by the current input. [Official reference](#).

Pitfall

“Dynamic” does not mean “unoptimized” and “static” does not mean “obsolete.” Modern PyTorch can compile eager programs, and modern TensorFlow supports eager execution. The useful question is which execution and compilation behavior fits the specific model and deployment target.

6 Automatic differentiation and backpropagation

Definition

A *gradient* collects partial derivatives of a scalar output with respect to one or more inputs. Automatic differentiation computes these derivatives by applying the chain rule to recorded operations. Backpropagation is the reverse traversal used to obtain gradients for neural-network parameters efficiently.

For the scalar function

$$z = w^\top x + b = w_1 x_1 + w_2 x_2 + b,$$

the gradient with respect to x is w , the gradient with respect to w is x , and the derivative with respect

to b is 1:

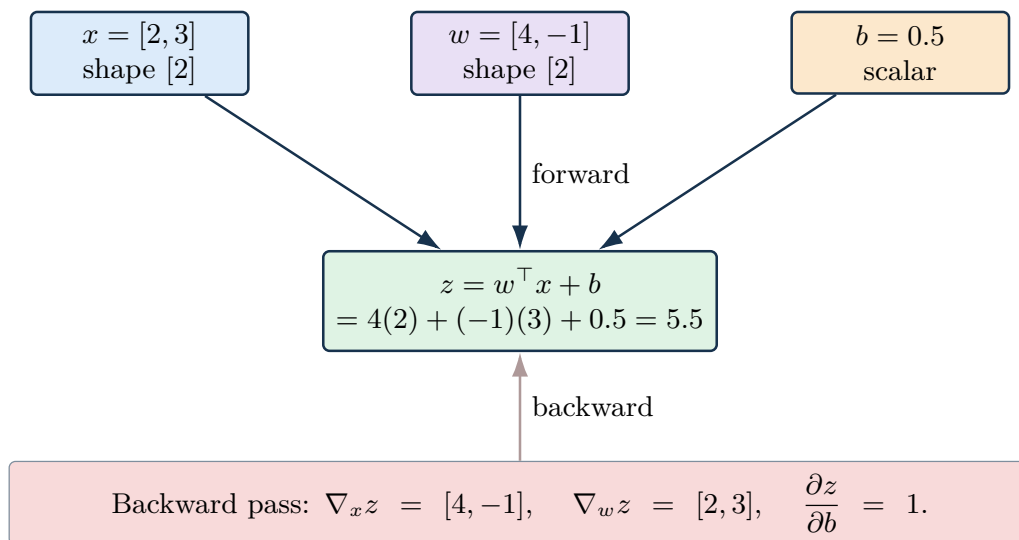
$$\nabla_x z = w, \quad \nabla_w z = x, \quad \frac{\partial z}{\partial b} = 1.$$

The symbol $\nabla_x z$ denotes the vector of derivatives of z with respect to every component of x .

With $x = [2, 3]$, $w = [4, -1]$, and $b = 0.5$,

$$z = 5.5, \quad \nabla_x z = [4, -1], \quad \nabla_w z = [2, 3], \quad \frac{\partial z}{\partial b} = 1.$$

Automatic differentiation on a scalar model



Autograd records differentiable operations and applies the chain rule in reverse to populate gradients.

Figure 6: The forward computation produces $z = 5.5$; the reverse pass produces gradients with shapes matching their corresponding inputs. [Official reference](#).

The following runnable example was executed with PyTorch and checked against the manual calculation.

```
import torch

x = torch.tensor([2.0, 3.0], requires_grad=True)
w = torch.tensor([4.0, -1.0], requires_grad=True)
b = torch.tensor(0.5, requires_grad=True)

z = torch.dot(w, x) + b
z.backward()

print(float(z.detach())) # 5.5
print(x.grad.tolist())   # [4.0, -1.0]
print(w.grad.tolist())   # [2.0, 3.0]
print(float(b.grad))     # 1.0
```

Listing 1: Build one scalar graph and compute its gradients

The state changes are precise. Before `backward()`, the tensors describe the forward calculation. After `backward()`, PyTorch has populated the `grad` fields of leaf tensors that requested gradients. Calling `backward()` repeatedly without clearing or replacing gradients accumulates values by design.

Training uses gradients to reduce a loss. A basic gradient-descent update is

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t).$$

Here θ_t is the parameter vector at optimization step t , $\mathcal{L}$ is a scalar loss, $\nabla_{\theta} \mathcal{L}$ is its gradient with respect to the parameters, and $\eta > 0$ is the learning rate. The equation moves parameters opposite the local direction of increasing loss.

7 Six core strengths

7.1 Efficient tensor computation

PyTorch provides vectorized operations, matrix multiplication, reductions, indexing, reshaping, convolution primitives, and many other tensor transformations. Vectorized code expresses a whole tensor operation rather than a Python loop over individual elements, allowing optimized kernels to perform the work.

7.2 Hardware acceleration

The same conceptual operation can run on different supported devices when tensors and parameters are placed consistently. A GPU is effective when the workload contains enough parallel numerical work to offset transfer and launch overhead. Merely installing a CUDA-capable build does not move data automatically.

Shape check

For an operation $z = Wx + b$, the tensors must agree in shape, dtype, and device. If W is on a CUDA device while x is on the CPU, the operation fails. If W expects $F = 128$ input features while x supplies $F = 127$, the matrix multiplication fails even when both tensors are on the same device.

7.3 Dynamic eager execution

Ordinary Python control flow supports rapid prototyping, data-dependent branches, recursive structures, and straightforward inspection. This property contributed strongly to early research adoption.

7.4 Automatic differentiation

`torch.autograd` removes the need to derive and implement every parameter gradient manually. It still requires mathematically valid differentiable operations and correct loss construction; autograd cannot repair an ill-defined objective or a detached graph.

7.5 Distributed training

When one device is insufficient, work can be distributed across several devices or machines. In data-parallel training, each worker processes a local mini-batch and gradients are synchronized before a consistent update. If worker r uses local batch size B_r , the global batch size is

$$B_{\text{global}} = \sum_{r=1}^R B_r,$$

where R is the number of workers. For four workers with $B_r = 32$, $B_{\text{global}} = 128$. Communication, data loading, failure recovery, and learning-rate behavior become additional design concerns.

7.6 Interoperability

PyTorch interoperates naturally with Python data tools and can export suitable programs to formats such as ONNX. Interoperability is never absolute: operators, dynamic shapes, control flow, precision, and runtime versions determine whether a particular model exports and runs correctly.

8 Research, optimization, saving, and deployment

Several operations that are often described together serve different purposes:

Eager execution runs Python model code directly and supports inspection and debugging.

Compilation attempts to optimize execution while preserving model behavior; `torch.compile` is the central PyTorch 2.x interface.

Saving parameters stores learned tensors, commonly through a `state_dict`; the model definition is still needed to reconstruct behavior.

Program export captures a deployable program representation under explicit assumptions about inputs and control flow.

ONNX export converts supported model behavior into the Open Neural Network Exchange representation for compatible runtimes.

Serving exposes model inference to callers through a process, service, or application-specific runtime.

Optimization, saving, and export answer different questions

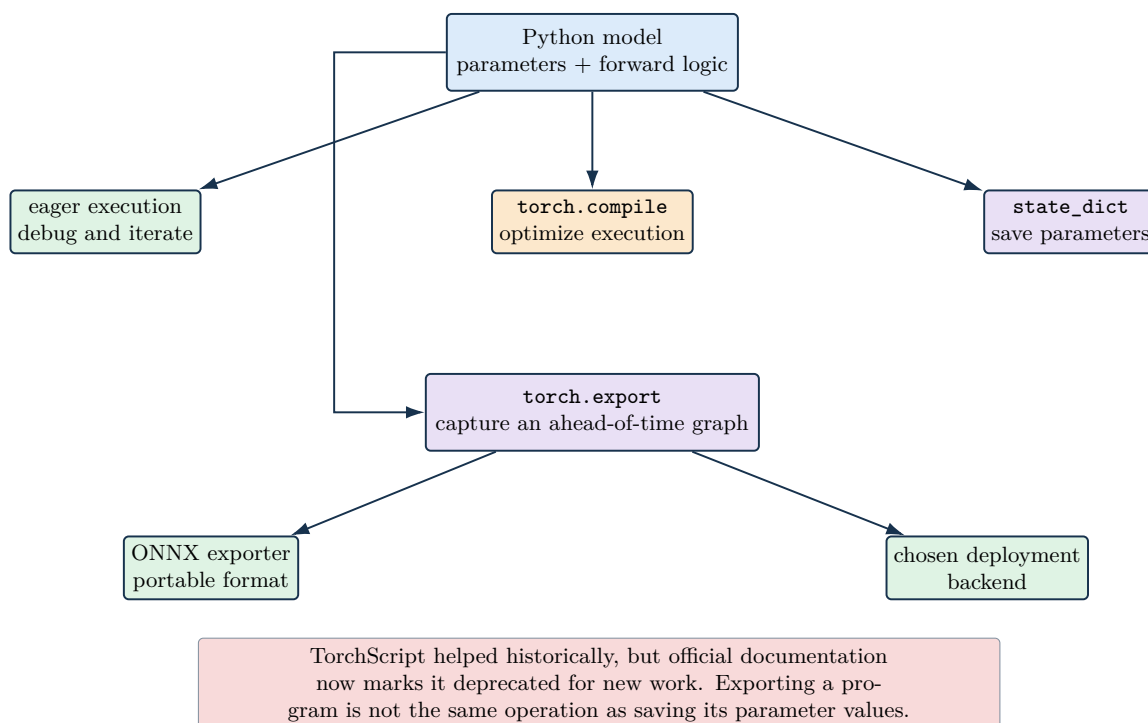


Figure 7: Eager execution, compilation, parameter saving, and program export are related but distinct workflows. [Official reference](#).

Current-practice note

TorchServe is described in older ecosystem material as a PyTorch model-serving solution. Its official documentation now carries a limited-maintenance notice stating that no updates, bug fixes, features, or security patches are planned. A new serving design should evaluate actively maintained alternatives and security requirements rather than selecting TorchServe solely from historical familiarity.

8.1 Quantization with corrected arithmetic

Quantization represents some values with lower precision to reduce payload size and potentially improve inference performance on supported hardware. If N_w weights each use q bits, the ideal raw payload is

$$S_{\text{bytes}} = N_w \frac{q}{8}.$$

The symbol S_{bytes} is payload size in bytes, N_w is the number of stored weights, and q is bits per weight. For one million weights:

$$S_{32} = 1,000,000 \times \frac{32}{8} = 4,000,000 \text{ bytes} \approx 4.0 \text{ MB},$$

$$S_8 = 1,000,000 \times \frac{8}{8} = 1,000,000 \text{ bytes} \approx 1.0 \text{ MB}.$$

The ideal payload reduction is $S_{32}/S_8 = 4$. A 400 MB versus 100 MB comparison would correspond approximately to *one hundred million* weights under the same simplified decimal calculation, not one million.

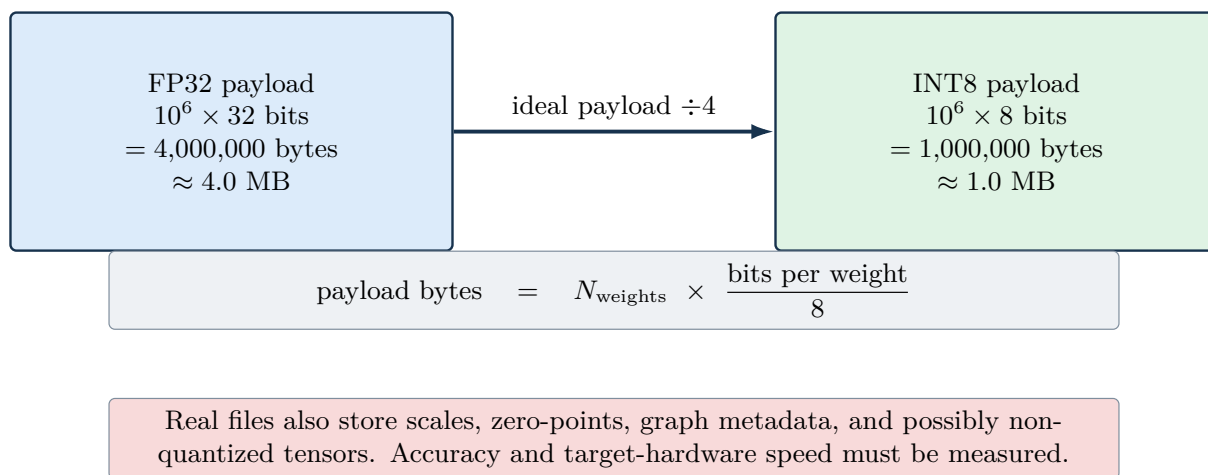
Quantization changes numeric representation

Figure 8: Reducing a one-million-weight payload from FP32 to INT8 gives an ideal fourfold payload reduction before metadata and other tensors. [Official reference](#).

Actual file size and speed differ from this idealized arithmetic because a quantized representation may store scales, zero-points, graph metadata, and non-quantized values. Quantization can also alter accuracy. The only reliable conclusion comes from measuring the exported model on the intended hardware and dataset.

9 Core modules and surrounding libraries

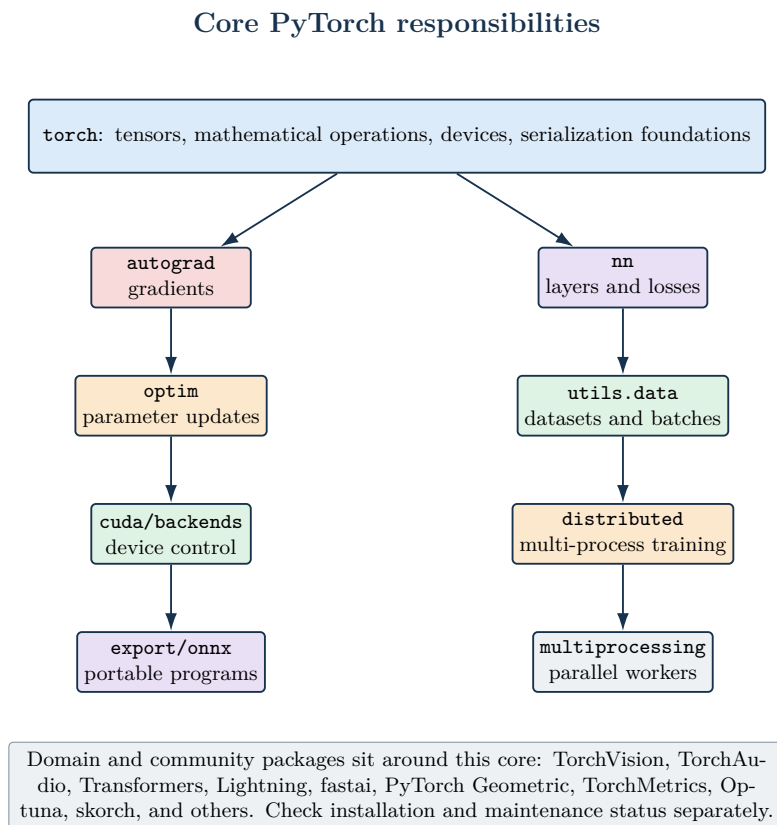


Figure 9: Core modules divide tensor mathematics, differentiation, modeling, optimization, data handling, device control, distribution, and export responsibilities. [Official reference](#).

9.1 Core modules

Modeling and training

Module	Responsibility
<code>torch</code>	Tensors, mathematical operations, device movement, random generation, serialization foundations, and the main namespace.
<code>torch.autograd</code>	Records differentiable operations and computes gradients by reverse-mode automatic differentiation.
<code>torch.nn</code>	Layers, activations, loss functions, containers, parameter registration, and the <code>Module</code> base class.
<code>torch.optim</code>	Optimization algorithms such as SGD, Adam, and RMSprop.
<code>torch.utils.data</code>	<code>Dataset</code> , <code>DataLoader</code> , samplers, batching, and data-pipeline utilities.
<code>torch.distributed</code>	Multi-process and multi-machine communication for distributed training.

Execution and portability

Module	Responsibility
<code>torch.cuda</code>	NVIDIA CUDA device interfaces, streams, events, memory tools, and CUDA availability checks.
<code>torch.backends</code>	Controls and reports backend behavior such as cuDNN or other platform-specific kernels.
<code>torch.multiprocessing</code>	Multiprocessing utilities that understand tensor sharing.
<code>torch.export</code>	Captures suitable PyTorch programs into an ahead-of-time graph representation.
<code>torch.onnx</code>	ONNX export and related interoperability tools.
<code>torch.jit</code>	Historical TorchScript and JIT functionality; deprecated for new export workflows in current official documentation.
<code>torch.ao.quantization</code>	Earlier in-tree quantization APIs; current development is moving toward <code>torchao</code> .

A module's presence does not guarantee that the current machine can use every feature. For example, `torch.cuda` can be imported in a CPU-only environment while `torch.cuda.is_available()` returns `false`.

9.2 Domain packages

Domain packages add data, models, and transformations for particular modalities:

- **TorchVision** provides computer-vision datasets, transformations, operators, and model architectures.
- **TorchAudio** has supplied audio I/O, transforms, datasets, and models; its current compatibility and maintenance direction should be checked before adoption.
- **TorchText** historically supplied NLP data utilities; many modern NLP workflows instead use tokenizers and datasets from the Hugging Face ecosystem.
- **TorchArrow** was presented as an experimental tabular and preprocessing project rather than a universal core dependency.
- **Lightning** provides a higher-level organization for training, evaluation, logging, and distributed execution while retaining PyTorch models underneath.

9.3 Community ecosystem

The ecosystem includes many independently maintained projects. Their responsibilities include:

Project	Typical purpose
Hugging Face Transformers	Pretrained transformer architectures and workflows for language, vision, audio, and multimodal tasks.
fastai	High-level training interfaces and practical defaults built on PyTorch.
PyTorch Geometric TorchMetrics	Graph neural networks and geometric deep learning. Reusable metric implementations with tensor and distributed-state handling.
TorchElastic / distributed elastic	Elastic and fault-aware distributed launching capabilities.
Optuna	Automated hyperparameter search and pruning.
Catalyst and Ignite	Higher-level training-loop engines and experiment organization.
AllenNLP	Research-oriented NLP components historically built on PyTorch.
skorch	A scikit-learn-compatible wrapper around PyTorch models.
PyTorch Forecasting	Higher-level time-series forecasting models and utilities.
TensorBoard integration	Visualization of metrics, graphs, and experiment information.

Pitfall

An ecosystem list is not an installation plan. Packages can be deprecated, renamed, archived, or incompatible with a chosen PyTorch version. Check official maintenance status, release dates, security posture, supported Python versions, and license before depending on any package.

10 PyTorch and TensorFlow: a constraint-based comparison

TensorFlow is another deep-learning framework whose early public design emphasized graph execution and production deployment. TensorFlow 2.x added eager execution and uses Keras as its principal high-level interface. The old slogan “PyTorch is dynamic, TensorFlow is static” is therefore incomplete.

10.1 Comparison dimensions

1. **Programming language and integration.** PyTorch is strongly Python-oriented and fits naturally into Python scientific workflows. TensorFlow has offered interfaces across a broader set of languages and environments. Verify the current support level of every non-Python interface.
2. **Ease of use and learning curve.** Many Python users find native PyTorch model code intuitive. TensorFlow’s Keras interface provides a mature high-level path. Ease depends on whether the task needs standard layers or low-level custom behavior.
3. **Customization.** Ordinary Python control flow and explicit `nn.Module` composition make custom PyTorch experimentation direct. TensorFlow also supports custom layers and operations, but the surrounding execution and export constraints differ.
4. **Performance.** Both frameworks use optimized kernels and compilers. A general winner cannot be inferred from dynamic versus static terminology; benchmark the target model, shapes, precision, and hardware.
5. **Deployment and production.** TensorFlow historically developed a broad serving, mobile, browser, and pipeline toolset. PyTorch expanded deployment through compilation, export, ONNX, and runtime integrations. Individual tools may have changed maintenance status.
6. **Mobile and embedded targets.** TensorFlow Lite and TensorFlow.js have been prominent in constrained-device and browser deployment. PyTorch-related mobile paths have evolved. Confirm that required operators, quantization modes, package size, and device runtimes are supported now.

7. **Community and preferred domains.** PyTorch gained early strength in research and later became central to many open generative-model workflows. TensorFlow developed a large industry ecosystem. These tendencies do not determine the best framework for every organization.
8. **High-level APIs.** Lightning and fastai add higher-level PyTorch workflows; Keras supplies TensorFlow’s established high-level interface. Extra abstraction reduces boilerplate but can hide state changes that a beginner should first understand explicitly.
9. **Interoperability.** PyTorch integrates well with Python tools and can export supported models to ONNX. TensorFlow provides SavedModel and other integration paths and can also participate in ONNX workflows. Operator support and version compatibility must be tested.
10. **Distributed training.** Both ecosystems support multi-device and multi-machine training. Usability and performance depend on launcher behavior, collective communication, fault tolerance, network topology, and observability.
11. **Pretrained models.** TorchVision, Hugging Face, TensorFlow Hub, and Kaggle Models provide large model collections. Model availability alone is insufficient; examine license, preprocessing, expected input shape, checkpoint format, and inference requirements.

Choose a framework from constraints, not a scoreboard

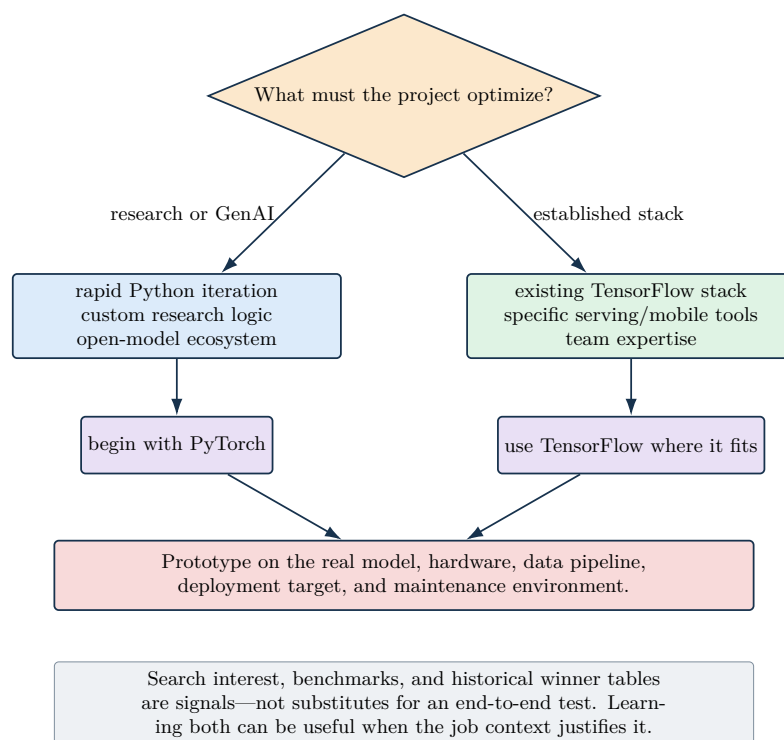


Figure 10: Framework choice should begin with project constraints and finish with a representative prototype on the intended system. [Official reference](#).

10.2 How to interpret popularity signals

A worldwide five-year search comparison presented PyTorch interest overtaking TensorFlow around the 2021–2022 period, while regional patterns differed. Such a chart is descriptive rather than causal. Search interest depends on query spelling, geography, time window, news cycles, education, and documentation usage; it does not directly measure production installations, job quality, or model performance.

A practical decision rule is:

- begin with PyTorch when the goal emphasizes Python experimentation, deep-learning fundamentals, generative AI, LLMs, or ecosystems whose reference implementations are PyTorch-first;
- use TensorFlow when an existing team, model fleet, mobile/browser target, or production platform already depends on TensorFlow-specific tooling;
- learn both when time and career context justify it, but master one complete workflow before collecting shallow familiarity with several frameworks.

11 Illustrative industry use cases

Large organizations use mixed technology stacks, so the examples below demonstrate possible PyTorch roles rather than exclusive company-wide commitments.

Organization	Products or areas cited	Illustrative PyTorch role
Meta	Facebook, Instagram, AI research	Computer vision, natural-language processing, and research; Meta's research organization originated PyTorch.
Microsoft	Azure Machine Learning, Bing, Microsoft 365 intelligent features	Cloud integration, search relevance, productivity tools, and AI development workflows.
Tesla	Autopilot and Full Self-Driving capability	Training perception and computer-vision models for autonomous-driving systems.
OpenAI	GPT-family systems, DALL-E, ChatGPT	Large-scale language and generative-model training.
Uber	Ride-hailing, Uber Eats, Pyro	Demand forecasting, route optimization, recommendation, and probabilistic programming through Pyro.

The transferable lesson is not the brand list. PyTorch can support research prototypes, large-scale training, computer vision, language modeling, forecasting, recommendations, probabilistic models, and deployed inference. Each production system still requires data engineering, evaluation, monitoring, security, cost control, and operational ownership beyond the framework itself.

12 A complete learning roadmap

A coherent progression through practical PyTorch is:

1. **Tensor fundamentals:** creation, shapes, dtypes, indexing, broadcasting, devices, and tensor mathematics.
2. **Computation graphs:** dependencies, eager execution, dynamic control flow, and graph breaks under compilation.
3. **Autograd:** gradients, leaf tensors, accumulation, chain rule, and disabling gradient recording for inference.
4. **Model definition:** `nn.Module`, parameters, layers, activations, and forward-shape contracts.
5. **Data pipelines:** `Dataset`, `DataLoader`, batching, transformations, sampling, collation, and workers.
6. **Training:** losses, optimizers, gradient clearing, forward pass, backward pass, update, metrics, and validation state.
7. **Architectures:** artificial neural networks, convolutional neural networks, recurrent neural networks, and LSTM-based sequence models.

8. **Efficiency:** GPU placement, mixed precision, profiling, compilation, input-pipeline performance, and distributed training.
9. **Model delivery:** parameter checkpoints, export constraints, ONNX where appropriate, quantization, runtime selection, and monitoring.

Tools change faster than mathematical foundations. A roadmap should therefore teach tensors, gradients, shapes, state, and optimization deeply, then attach current deployment APIs after checking official documentation.

13 Failure modes and diagnostic checks

1. **Undefined shape assumptions:** write expected shapes next to every input, output, label, and parameter tensor.
2. **Wrong dtype:** integer token IDs, floating-point features, and class targets serve different API contracts.
3. **Device mismatch:** move both model state and input tensors to compatible devices; check availability rather than assuming a GPU is usable.
4. **Detached computation:** converting a needed tensor to a non-differentiable representation or explicitly detaching it breaks the gradient path.
5. **Gradient accumulation:** gradients accumulate across backward calls unless cleared or replaced deliberately.
6. **Confusing graph and architecture:** a box-and-arrow network diagram describes model structure; an autograd graph records actual tensor operations for a particular execution.
7. **Assuming compilation always helps:** compilation has capture and startup costs and can encounter graph breaks; measure steady-state behavior.
8. **Confusing saving with export:** a parameter checkpoint does not contain an independent inference program unless the chosen format explicitly captures one.
9. **Treating ONNX as automatic portability:** unsupported operators, dynamic control flow, shape constraints, and runtime differences can block or alter export.
10. **Incorrect quantization arithmetic:** always convert bits to bytes and distinguish decimal MB from binary MiB.
11. **Ignoring quantization accuracy:** smaller representation does not guarantee acceptable predictions or speed on the target processor.
12. **Assuming more devices are faster:** communication and input bottlenecks can dominate distributed training.
13. **Following stale ecosystem guidance:** verify deprecation and maintenance notices for TorchScript, TorchServe, quantization APIs, and domain packages.
14. **Choosing from popularity alone:** search trends and community size do not replace a representative end-to-end prototype.

14 Key takeaways and review questions

Key takeaways

- PyTorch unifies tensors, differentiable operations, neural-network components, optimization, hardware execution, distribution, and export.
- Shape, dtype, and device are first-class tensor contracts.
- Dynamic eager execution records operations as Python runs; automatic differentiation traverses those dependencies in reverse.
- PyTorch’s evolution reflects a continuing effort to combine research flexibility with efficient and portable execution.
- Historical tools remain useful context, but current official guidance deprecates TorchScript for new export work, centralizes quantization work in `torchao`, and marks TorchServe as limited maintenance.
- Framework choice should follow model, team, hardware, deployment, and maintenance constraints rather than a universal winner table.

Review questions

1. What three properties, besides numerical values, define a tensor’s operational contract?
2. Why does an image shape $[B, C, H, W]$ differ semantically from $[B, H, W, C]$?
3. Expand $y = (a + b)c$ into intermediate operations for $a = 1$, $b = 2$, and $c = 5$.
4. What information does a computation-graph edge represent?
5. How does dynamic eager execution handle a data-dependent branch?
6. For $z = w^\top x + b$, why is $\nabla_w z = x$?
7. What state changes when `backward()` is called?
8. Why must ordinary gradients be cleared between optimization steps?
9. Distinguish eager execution, compilation, saving a `state_dict`, and exporting a program.
10. Compute the ideal payload for 25 million FP32 weights and for the same count stored as INT8.
11. Why might the real quantized file be larger than the ideal INT8 payload?
12. What does `torch.utils.data` provide that `torch.nn` does not?
13. Why can `torch.cuda` exist even when CUDA execution is unavailable?
14. Which project constraints would make an existing TensorFlow stack preferable to a migration?
15. Why is a Google Trends curve insufficient evidence for framework performance or production adoption?